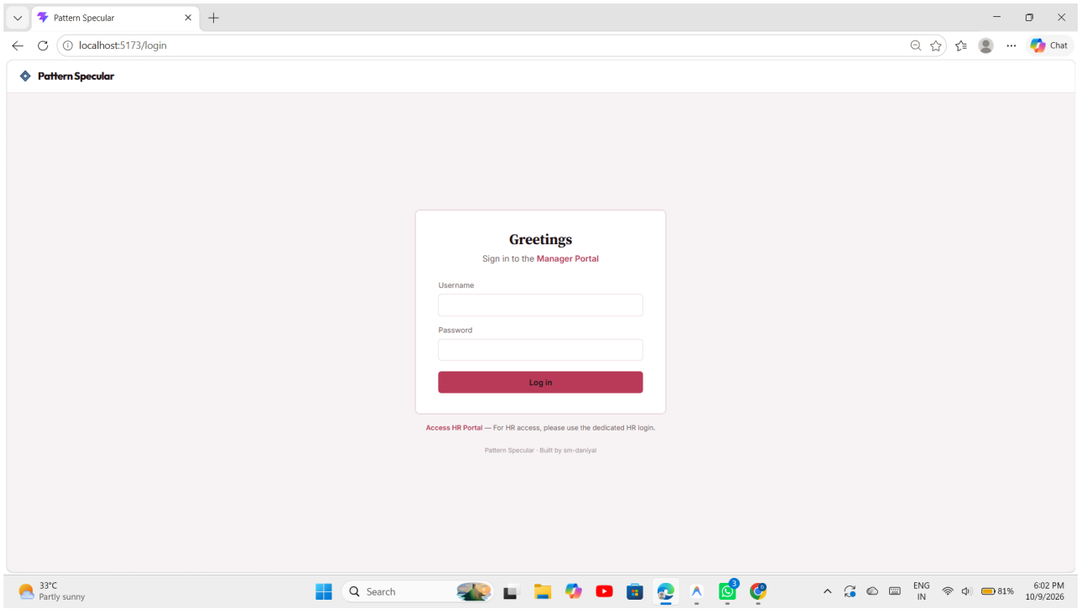
◆ PATTERN SPECULAR

The Pattern Specular

A Longitudinal Bias-Pattern Analysis Platform for Talent Writing

"Surfaces bias patterns in a manager's own writing, with statutory and academic evidence, so they can decide what to change."

Author: sm-daniyal (Syed Daniyal)
Stack: React 19 · FastAPI · Google Gemini · Neon Serverless PostgreSQL



About the Author

Full-Stack Software Engineer & Applied-AI Practitioner

Syed Daniyal (sm-daniyal)

GitHub: github.com/sm-daniyal/patternSpecular · **Location:** India

Core Competencies:

- **Full-Stack Engineering:** Modern React 19 single-page apps, TanStack Router/Query, Vite, Tailwind CSS tokens.
- **Backend Systems:** Python 3.12, FastAPI asynchronous APIs, SQLAlchemy 2.0, robust state machines, SSE streaming.
- **Applied AI & LLM Systems:** Google Gemini 2.5/3.5, schema-first structured outputs with Instructor, multi-agent debate loops.
- **Statistical Data Science:** Finite-sample exact tests (Fisher's exact $p < 0.05$), differential privacy ($k \ge 3$).

Project Motivation:

Corporate DEI training leaves a major visibility gap: managers receive unconscious bias training, yet no tool provides continuous, private calibration over the documents they actually write. Pattern Specular was engineered as an autonomous mirror — evidence-anchored, mathematically rigorous, and structurally private.

Walkthrough Agenda

Three Sections across 17 Presentation Slides

SECTION 1: THE PRODUCT

04 Motivation: Where Training Stops, This Begins

05 What It Does: Evidence In, Decision Theirs

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16 Production Engineering Learnings

17 Repository & Contact

Motivation — Where Bias Training Stops, This Begins

Why corporate fair-hiring theory fails to change real hiring decisions

THE CRITICAL VISIBILITY GAP

Enterprises spend millions on annual fair-hiring and unconscious bias training. However, once training ends, **no continuous system verifies or assists** the actual writing produced by managers. Writing is where hiring, evaluation, and promotion decisions are codified.

BIAS IS A LONGITUDINAL PATTERN

In isolation, a single evaluation containing phrases like *'warm and polite'* or *'aggressive go-getter'* appears completely harmless. Subtle bias only reveals itself as a statistically significant pattern across **20+ documents evaluated over time**.

THE PATTERN SPECULAR RESPONSE

Pattern Specular acts as an autonomous calibration mirror: it flags statutory violations in real time, verifies candidate feedback against the job description's stated criteria, and surfaces longitudinal tendencies using Fisher's Exact Test ($p < 0.05$) — **never as a punitive score**.

What It Does — Evidence In, Decision Theirs

A calibrated mirror for writers, preserving 100% human autonomy

Core Product Capabilities:

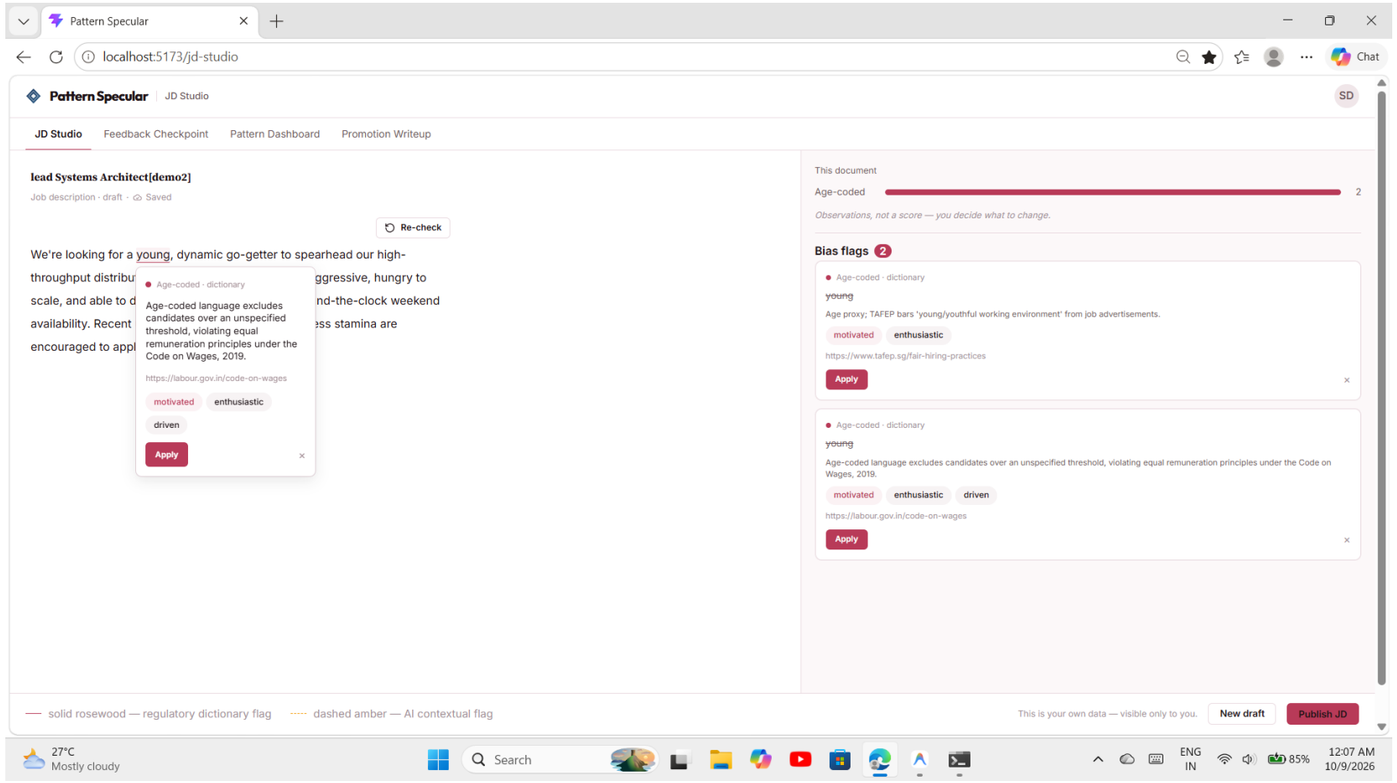
- **Every Flag is Cited:** Cites labor legislation (India Code on Wages 2019, RPwD Act 2016; Singapore TAFEP; US EEOC; UK Equality Act 2010) or peer-reviewed research. Unbacked flags are automatically pruned.
- **Criteria Drift Protection:** Flags when interview feedback fixates on subjective tone while ignoring stated technical skills.
- **Longitudinal Significance:** Only surfaces tendencies that clear Fisher's Exact Test ($p < 0.05$) across months.
- **Autonomous Self-Growth:** Autonomous 4-agent debate proposes dictionary updates from emerging workplace slang.

Non-Negotiable Design Invariants:

- **Mirror, Not Judge:** Never scores, grades, or penalizes managers.
- **Private by Architecture:** Manager writing is structurally unreachable by HR queries.
- **Non-Blocking:** Every flag is dismissible; managers retain complete hiring authority.
- **Not a Chatbot:** No conversational fluff — deterministic and structured analysis only.
- **Not a Hiring Gate:** Assumes manager competence and good intent.

Manager Surfaces — JD Studio & Real-Time Feedback

Dual-pass detection: sub-10ms dictionary lemmatization + async LLM contextual pass



Surface 1: JD Studio

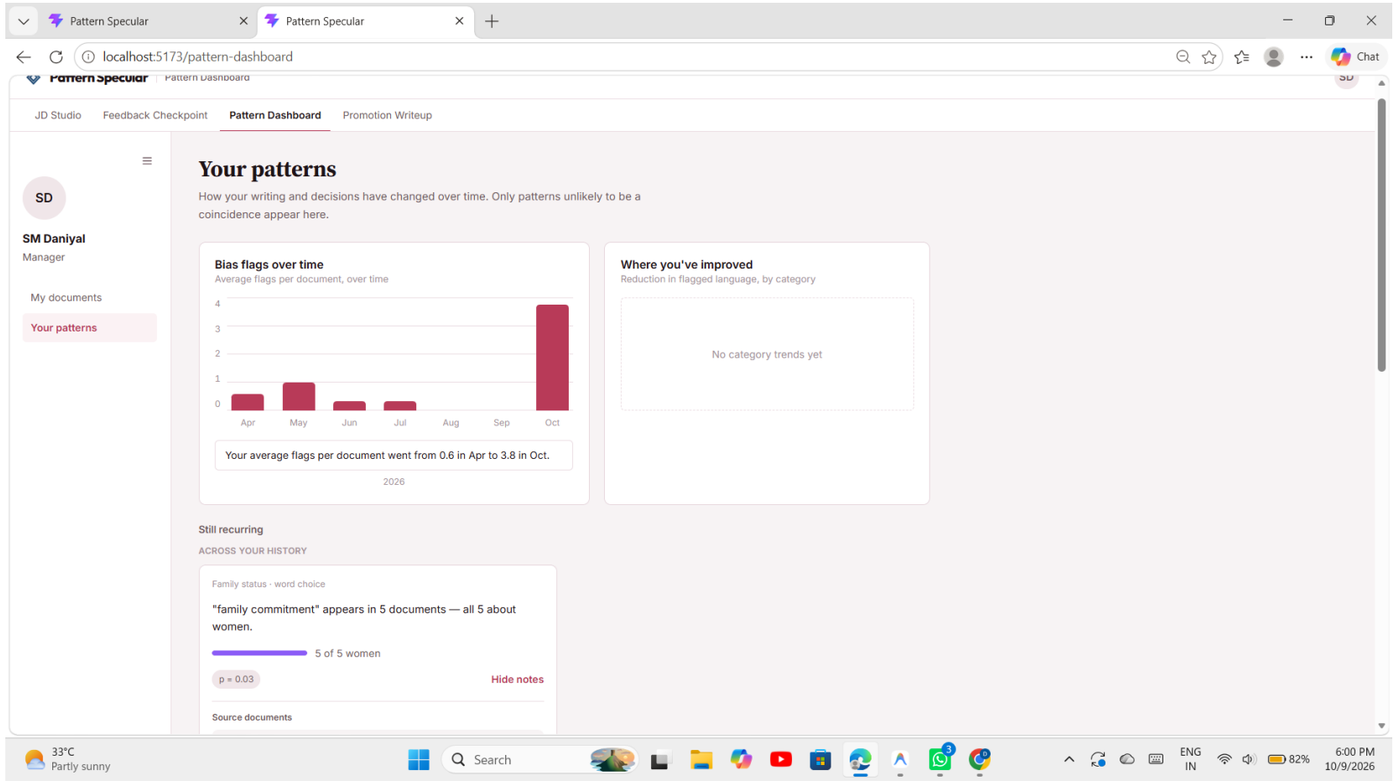
- **Instant spaCy Lemma Pass (<10ms):** Deterministically highlights age-coded keywords (`young`, `fresh`, `digital native`) against statutory lexicons.
- **Async Contextual Pass (Gemini):** Evaluates surrounding sentence semantics on typing pause to check for genuine occupational justifications.
- **Rosewood Statutory Card:** Cites labor legislation directly (India Code on Wages 2019, TAFEP) with one-click neutral alternatives (`motivated`, `driven`).

Surface 2: Feedback Checkpoint

Detects gender-coded evaluation language (`warm`, `supportive`, `aggressive`) while running **Criteria Drift Check** against the JD's required technical competencies.

Manager Surfaces — The Longitudinal Pattern Dashboard

A habit is a statistical finding ($p < 0.05$), never an impression or raw count

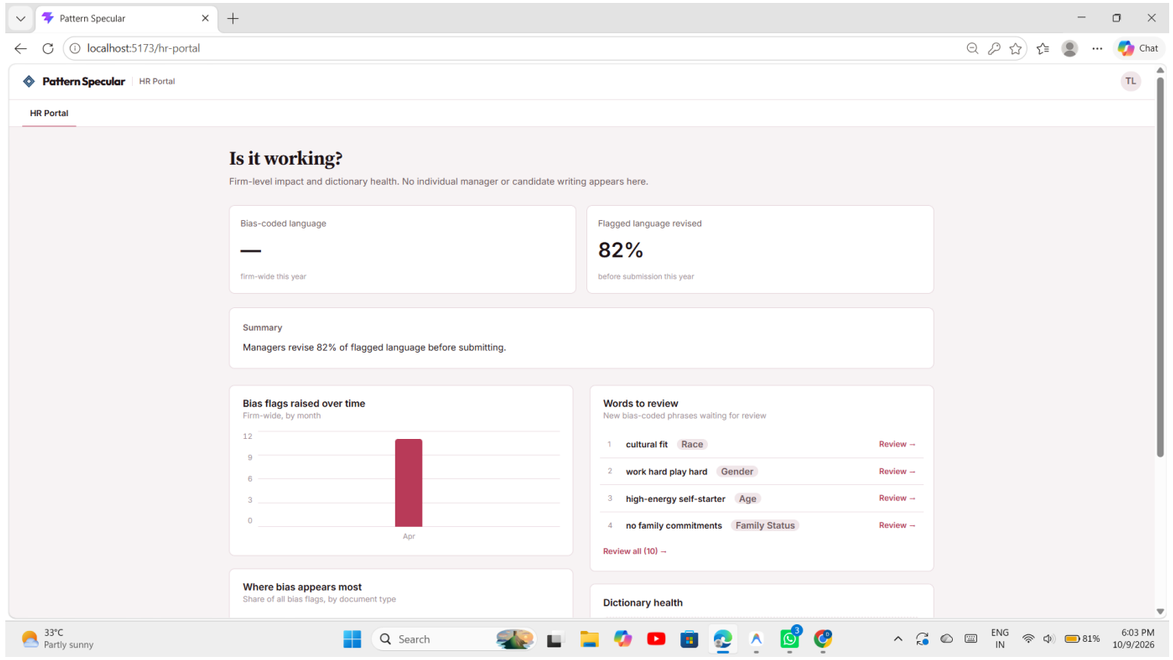


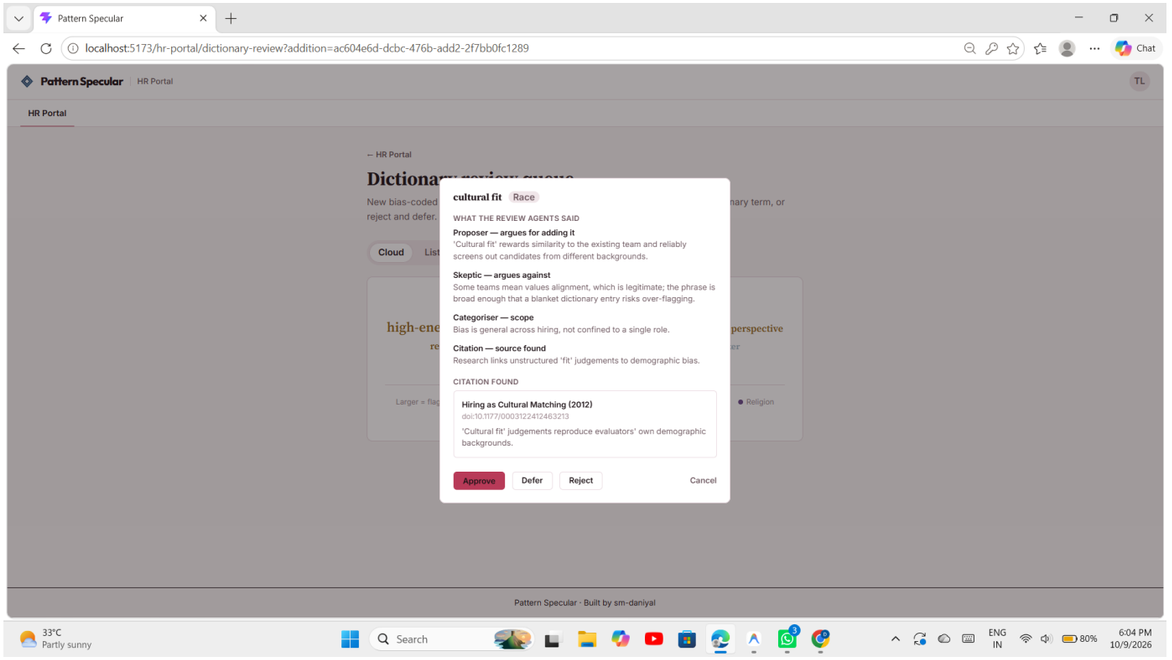
Statistical Rigor over Raw Counts:

- **12-Month Longitudinal Bar Charts:** Displays monthly flag density across age, gender, and family status categories.
- **Fisher's Exact Test Gating:** The card *'family commitment appears in 5 documents — all 5 about women'* is surfaced because the 2×2 contingency matrix clears **$p = 0.03 < 0.05$**.
- **Differential Adoption Tracking:** Separates flags the manager revised vs dismissed, showing upward behavioral learning slopes.
- **Private to Manager:** Backed by strict database `owner_id` tenancy isolation.

HR View — Firm-Wide Trends & Structural Anonymity

Firm-level metrics with mathematical $k \ge 3$ cell suppression





Left: Firm-Wide Effectiveness Portal — 82% revision rate, aggregate monthly volume, and document type distributions. **Right: Autonomous Dictionary Review** — 4-agent consensus debate modal for proposed term additions (Proposer, Skeptic, Categoriser, Citation).

Differential Privacy Invariant ($k \ge 3$): Any department or filter with fewer than 3 distinct managers is mathematically suppressed from HR views to prevent individual de-anonymization.

Engineering — The 7-Stage Hybrid Engine

Deterministic verification guarantees zero LLM hallucinations reaching the user

Stage 1: Deterministic Pass

spaCy regex & lemmatizer (<10ms). Instant exact matches against region-scoped statutory lexicons.

Stage 2: Contextual Pass

Async Google Gemini 3.5 Flash Lite call. Evaluates coded language, agentic/communal adjectives, and GDOR exceptions.

Stage 3: Adjudicator

Strict Python span verification. Confirms verbatim existence in original document; hallucinated spans dropped.

Stage 4: Verdict Module

Applies statutory rulings and bona-fide job requirements. Evaluates candidate justifications in context.

Stage 5: Suppression Module

Computes sentence-level SHA-256 fingerprint hashes against historical manager dismissals.

Stage 6: Judge Agent

Consensus multi-sampling confidence evaluation for borderline flags (threshold: ≥ 0.70).

Stage 7: Recommendations

Generates 2–3 evidence-grounded neutral alternative rewrite chips.

Engineering — LLM as a Judge (Consensus Confidence)

Adversarial second-opinion mechanism eliminating generator bias

1. Independent Model Architecture

To mitigate LLM self-preference bias, the Judge stage evaluates surviving candidate flags independently, without seeing the generator's explanation or internal chain-of-thought.

2. Rubric Checklist Decomposition

Rather than asking the model for an arbitrary 1–5 score, the judge evaluates discrete boolean criteria: `references_characteristic`, `gdor_plausible`, and `stated_objectively`.

3. Frequency-Based Confidence

Confidence is computed as empirical agreement frequency over N independent stochastic samples. Flags with agreement $< 70\%$ are discarded.

4. Position Invariant Shuffling

Candidate flag order is randomized prior to prompt serialization to eliminate list-position and primacy bias.

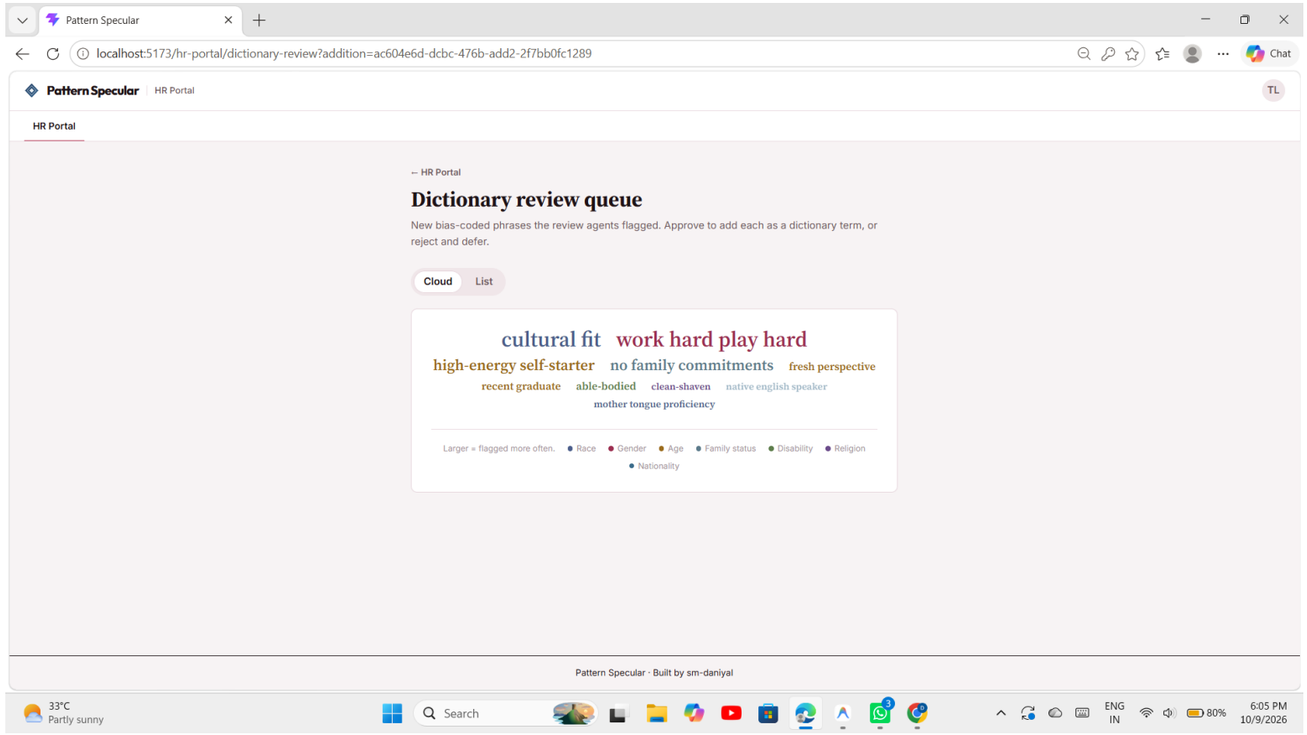
Engineering — Published ML Research Foundations

Five landmark papers grounding Pattern Specular's design choices

Paper / Authors	Year & Venue	Architectural Pattern Ingested
Panickssery, Bowman & Feng	2024 · arXiv:2404.13076	Different-model Judge architecture to avoid causal LLM self-preference.
CheckEval (Lee et al. / Gu et al.)	EMNLP 2025 · arXiv:2403.18771	Rubric decomposition into discrete boolean sub-questions over holistic scoring.
Wang et al. / Xiong et al.	ICLR 2023 · arXiv:2203.11171	Self-consistency via majority voting across independent generation paths.
Wang et al. / Zheng et al.	NeurIPS 2023 · arXiv:2306.05685	Randomized list permutation to counter position bias in LLM evaluators.
Tian et al. / Feuer et al.	EMNLP 2023 · arXiv:2603.05485	Empirical calibration curves for structured probability thresholds.

Engineering — Autonomous Dictionary Growth Loop

Self-evolving statutory lexicon through 4-agent debate



The 4-Agent Debate Committee:

- **Proposer Agent:** Identifies recurring phrases ($\\ge 2$ docs) and argues why it introduces protected-class exclusion.
- **Skeptic Agent:** Tests whether the phrase has valid, non-discriminatory business meaning in tech.
- **Categoriser Agent:** Scopes whether the term is globally applicable or role-specific.
- **Citation Agent:** Verifies labor statutes (India, SG, US, UK) or scholarly research.

Human-in-the-Loop Sign-Off:

Approved terms are added to the regional dictionary, converting future occurrences into **zero-latency (<10ms) deterministic Stage 1 hits**.

Engineering — System Architecture & Tech Stack

Modern decoupled cloud-native topology

FRONTEND CLIENT

- **React 19 & Vite:** High-performance SPA with concurrent rendering.
- **TipTap / ProseMirror:** Real-time rich text editor with custom decoration marks.
- **Tailwind CSS v4:** Atomic Rosewood token system (`#b83a58`).
- **TanStack Router & Query:** Fully type-safe client-side state caching.

BACKEND SERVICES

- **Python 3.12 & FastAPI:** High-throughput async REST endpoints & SSE.
- **LangGraph & State Machine:** Deterministic state transitions across 7 stages.
- **Structlog:** Structured JSON logging with trace correlation IDs.
- **scipy.stats:** Exact finite-sample hypergeometric distribution solver.

DATA & AI LAYER

- **Neon Serverless PostgreSQL 16:** Cloud-pooled branchable database with connection pooling.
- **SQLAlchemy 2.0 & Alembic:** 20 versioned schema migrations.
- **Google Gemini 3.5 Flash Lite:** Cost-efficient structured reasoning via Instructor.

Engineering — Data Model & Structural Privacy

Privacy guaranteed at the database schema layer, not by application policy

17 Core Relational Entities

- `User` & `UserRoleAssignment` (Manager vs HR role separation)
- `Document` & `AnalysisRun` (Immutable audit versions)
- `Flag`, `FlagDismissal`, `FlagInteraction` (Decision telemetry)
- `Dictionary` & `Citation` (Mandatory non-nullable statutory FK)
- `DriftFinding` & `PromotionRubricCriterion` (JD / Rubric grounding)
- `AgentRun` (Comprehensive LLM cost & prompt audit trail)

Structural Privacy Guarantees

- **Strict Tenant Queries:** Every manager read is parameterized by `owner_id`. No endpoint exists allowing a manager to query another's documents.
- **HR Structural Isolation:** HR aggregate models explicitly exclude document text fields. It is architecturally impossible for HR to read drafts.
- **Immutable Audit Trail:** Every LLM prompt, token count, and latency metric is logged in `AgentRun` for compliance audits.

Wrap-up — Version 2 Architectural Seams

Engineered for modularity and seamless enterprise extensions

1. Multi-Model Runtime Gateway

The `StructuredCompletionClient` protocol abstracts LLM providers. V2 enables runtime dynamic routing across Google Gemini, Claude, and local models.

2. Multimodal Video Drift Reference

Ingesting video interview recordings directly into Gemini's native audio/visual context window to verify spoken candidate claims against resume qualifications.

3. Historical Semantic Search (pgvector)

Enabling managers to query past evaluations with semantic embeddings, identifying recurring language across entire promotion cohorts.

4. Global Statutory Plug-ins

Adding new jurisdiction modules (EU AI Act, Australian Fair Work Act) by creating region-scoped citation tables without touching the engine core.

Wrap-up — Engineering Learnings & Production Standards

Key architectural lessons from building Pattern Specular

1. Rigorous Test Automation

240 backend tests (Pytest) and 251 frontend tests (Vitest) run across CI with 100% pass rates, verifying every state transition and statistical calculation.

2. Schema-First Contracts

Combining Pydantic v2 with Instructor guarantees 100% schema validation across AI boundaries, eliminating malformed JSON production crashes.

3. Privacy Enforced by Schema

Enforcing privacy invariants in database schemas rather than relying on application authorization checks prevents data leakage bugs by design.

4. Mathematical Gating

Preventing alert fatigue by requiring statistical significance ($p < 0.05$) ensures managers trust surfaced findings rather than dismissing noisy flags.

Project Repository & Contact

Pattern Specular is an open source sovereign portfolio project

Thank You!

GitHub Repository: https://github.com/sm-daniyal/patternSpecular

Author / Maintainer: sm-daniyal (Syed Daniyal)

Documentation Suite:

- `docs/DESIGN_SPEC.md` — Complete 14-section system design specification.
- `docs/STUDY_GUIDE.md` — 10-phase engineering mastery guide covering pipelines, statistical calibration, and multi-agent loops.
- `docs/adr/` — 18 Architecture Decision Records documenting all major technical decisions.
- `docs/demo/DEMO_FLOW.md` — 10-minute live recruiter presentation script.

Designed, implemented, verified, and calibrated by sm-daniyal.